



Machine Learning

CMPE 257

Session 2

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Agenda

- Last week recap
- Logistic Regression
- Generalized Linear Model (GLM)
- Constructing GLMs
- Naïve Bayes

Last Week Recap

Regression vs. Classification

Regression

- **Output type:** Continuous value
- **Goal:** Predict a numeric quantity
- Examples:
 - Predicting house prices
 - Forecasting temperature
- **Model types:** Linear regression, Ridge/Lasso regression

features/input	label/output
$x \in \mathbb{R}^2$	$y \in \mathbb{R}$

Classification

- **Output type:** Discrete label
- **Goal:** Assign input to a category
- Examples:
 - Spam vs. not spam
 - Diagnosing a disease (positive/negative)
- **Model types:** Logistic regression, SVM, Decision trees

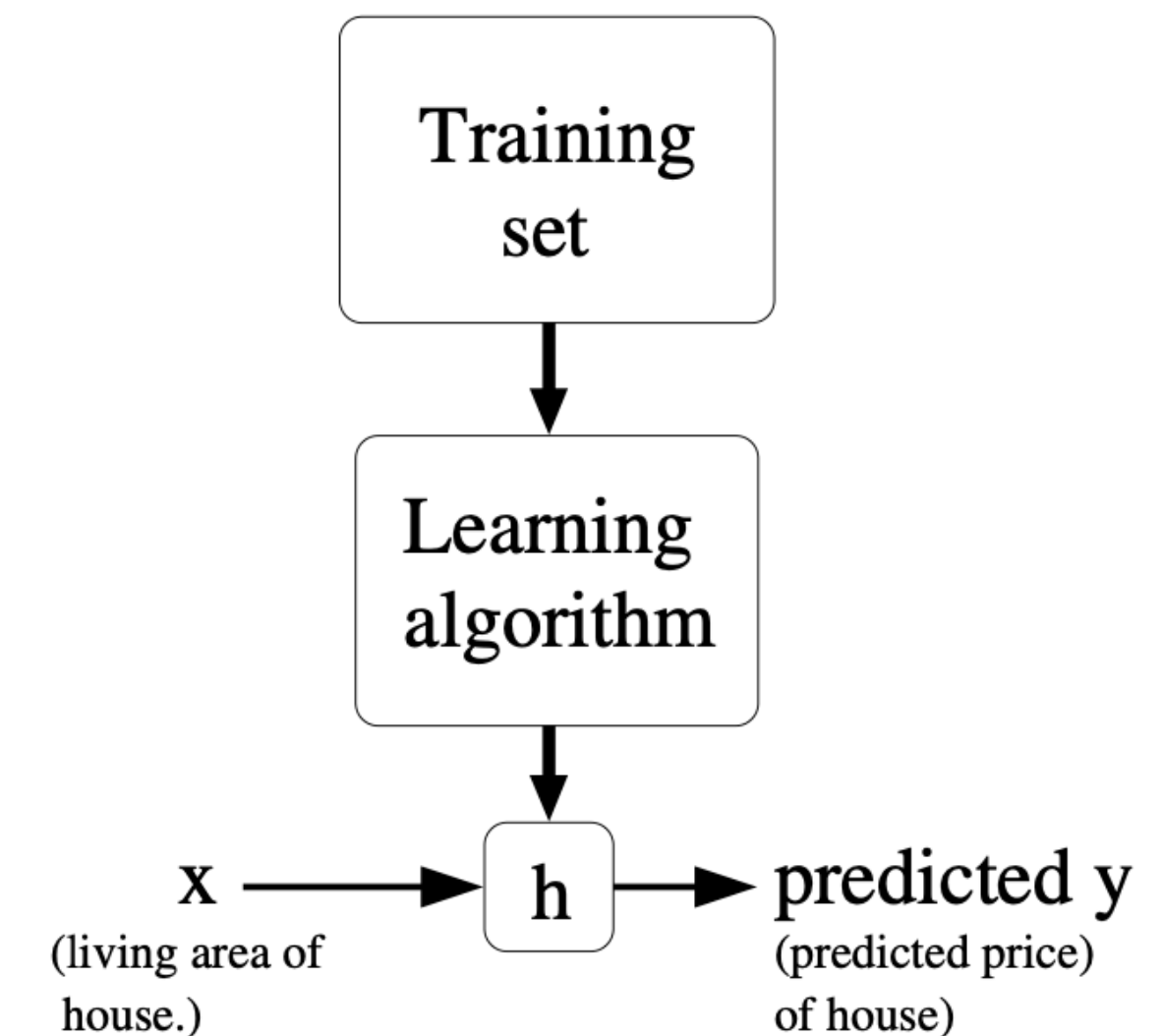
Key Difference

- **Regression** → Predict “How much?”
- **Classification** → Predict “Which class?”

Supervised Learning

Formal Definition

- Given a training set, to learn a function $\hat{h} : \mathcal{X} \rightarrow \mathcal{Y}$ so that $\hat{h}(x)$ is a “good” predictor for the corresponding value of y .
- function $\hat{h}$ is called a **hypothesis**.
- **Hypothesis** : $\hat{h}_{\mathcal{I}}(x) = \mathcal{I}^T x$
 - Parameters: $\mathcal{I}$
 - Features: x
- **Regression**: When the target variable that we’re trying to predict is continuous, such as in our housing example, we call the learning problem a regression problem.
 - Example: Predict house prices ($y \in \mathbb{R}$).
- **Classification**: When y can take on only a small number of discrete values (such as if, given the living area, we wanted to predict if a dwelling is a house or an apartment, say), we call it a classification problem.
 - Example: Spam vs. Non spam ($y \in \{0,1\}$).



Linear Regression

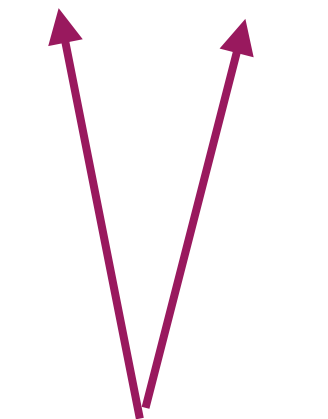
Hypothesis Function

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

Goal: Here, the θ_i 's are the **parameters** (also called **weights**) parameterizing the space of linear functions mapping from $\mathcal{X}$ to $\mathcal{Y}$.

$x_0 = 1$ this is the intercept term

d : number of input variables

$$h(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$$


Vectors

LMS — Least Mean Squares

Batch Gradient Descent on the cost function J

- For a training set of more than one example

Repeat until convergence {

$$\theta_j := \theta_j + \alpha \sum_{i=1}^n (y^{(i)} - h_{\theta}(x^{(i)})) x_j^{(i)}, (\text{for every } j) \quad (1)$$

}

- Or, an update of the vector θ

$$\theta := \theta + \alpha \sum_{i=1}^n (y^{(i)} - h_{\theta}(x^{(i)})) x^{(i)}$$

Supervised Learning

Logistic Regression

Supervised Learning — Classification

Logistic Regression

- Input $\rightarrow$ Output mapping from labeled data
- The values y , we now want to predict, take on only a small number of discrete values.
- We will focus on the **binary classification** problem in which y can take on only two values, 0 and 1.
- Notation:
 - $x^{(i)}$: input variable; features
 - $y^{(i)}$: class label; label, something that we want to predict; $y \in \{0, 1\}$
 - A pair $(x^{(i)}, y^{(i)})$ is called training example
 - *Training set*: a list of n training examples $\{(x^{(i)}, y^{(i)}); i = 1, \dots, n; n \in \mathbb{R}\}$
 - The superscript “ (i) ” in the notation is simply an index into the training set
 - We use $\mathcal{X}$ denote the space of input values, and $\mathcal{Y}$ the space of output values. In this example, $\mathcal{X} = \mathcal{Y} = \mathbb{R}$.

Logistic Regression

Formal Definition

- Given a training set, to learn a function $\hat{h} : \mathcal{X} \rightarrow \mathbf{Y}$ so that $\hat{h}(x)$ is a “good” predictor for the corresponding value of y .

- function $\hat{h}$ is called a **hypothesis**.

- Hypothesis** : $\hat{h}_{\vartheta}(x) = \vartheta^T x$

- Parameters: ϑ
$$h_{\theta}(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}},$$

- Features: x

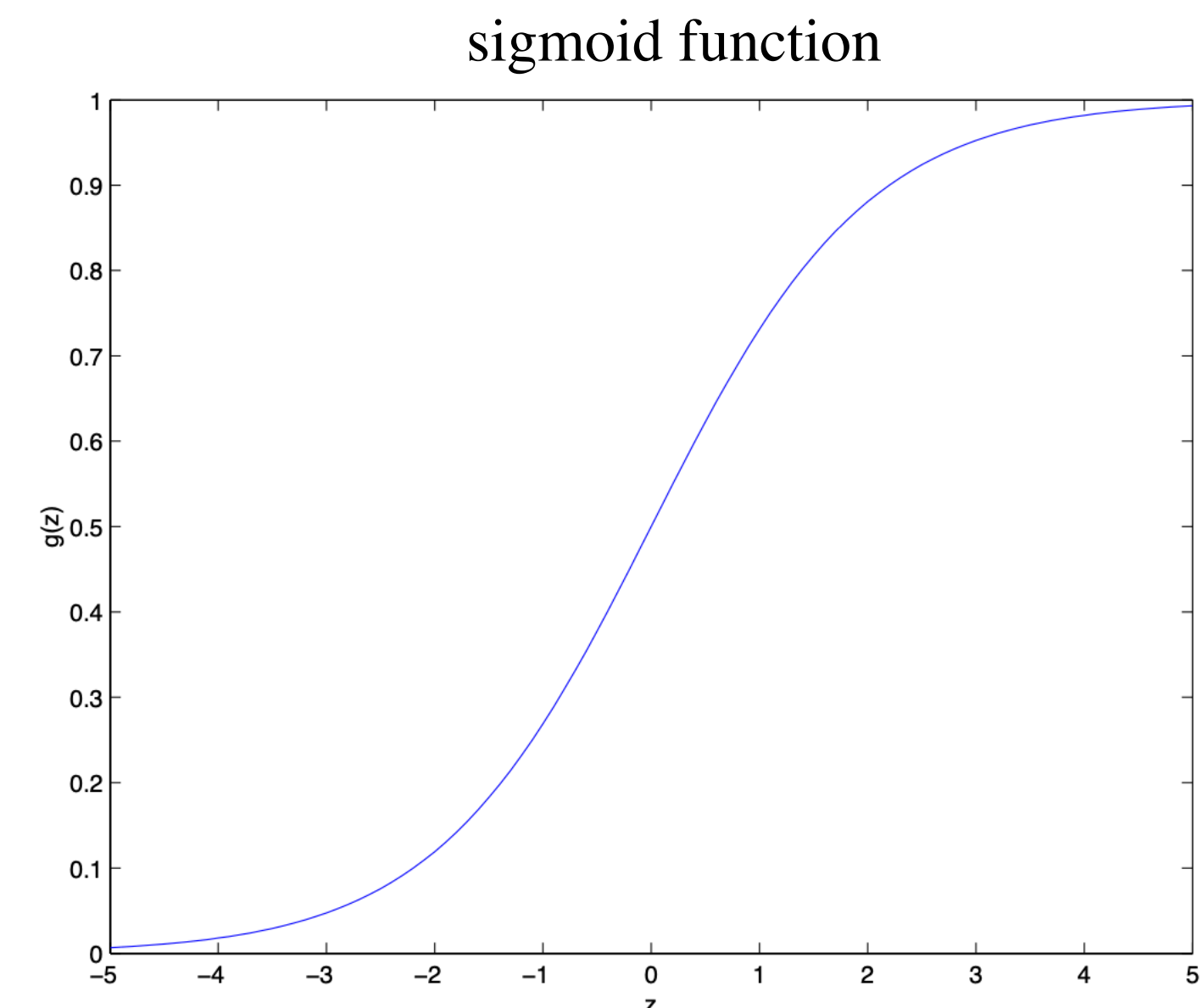
- Logistic Function or Sigmoid function**

- $g(z)$ tends towards 1 as $z \rightarrow \infty$, and
$$g(z) = \frac{1}{1 + e^{-z}}$$

- $g(z)$ tends towards 0 as $z \rightarrow -\infty$.

- $g(z)$, and hence also $\hat{h}(x)$, is always bounded between 0 and 1.

- Other functions that smoothly increase from 0 to 1 can also be used



Logistic Regression

Logistic Function

A useful property of the derivative of the sigmoid function, which we write as g :

$$\begin{aligned} g'(z) &= \frac{d}{dz} \frac{1}{1 + e^{-z}} \\ &= \frac{1}{(1 + e^{-z})^2} (e^{-z}) \\ &= \frac{1}{(1 + e^{-z})} \cdot \left(1 - \frac{1}{(1 + e^{-z})} \right) \\ &= g(z)(1 - g(z)). \end{aligned}$$

Logistic Regression

Modeling

Given the logistic regression model, how do we fit θ for it?

$$\theta^T x = \theta_0 + \sum_{j=1}^n \theta_j x_j$$

Logistic Regression

Maximum Likelihood Estimation

In **logistic regression**, we want to find the best parameters (θ) that make our model predict correctly whether something belongs to class 0 or 1.

In **linear regression**, we used **least squares** (minimizing squared errors). But for classification, squared errors don't really make sense because we are looking at class labels.

Instead, we look at it from a **probability perspective**:

- Logistic regression assumes the probability of being in class 1 follows the **sigmoid function** (an S-shaped curve).
- For each data point, the model gives a probability of it being 0 or 1.
- Then we ask: If our model is correct, what's the probability of observing the data we actually have?
 - This is called the **likelihood**.
- Finally, we choose the θ values that **maximize this likelihood** — meaning the parameters that make the observed data most probable.
 - This is the **Maximum Likelihood Estimation (MLE)** approach.

Logistic Regression

MLE

- Lets us assume that for a binary classification:

$$\begin{aligned}P(y = 1 \mid x; \theta) &= h_{\theta}(x) \\P(y = 0 \mid x; \theta) &= 1 - h_{\theta}(x)\end{aligned}$$

- This can be summarized as:

$$p(y \mid x; \theta) = (h_{\theta}(x))^y (1 - h_{\theta}(x))^{1-y}$$

Logistic Regression

MLE

- This can be summarized as: $p(y \mid x; \theta) = (h_\theta(x))^y (1 - h_\theta(x))^{1-y}$
- Assuming that the m training examples were generated independently, we can then write down the likelihood of the parameters as

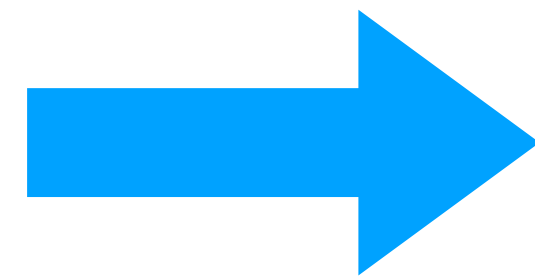
$$\begin{aligned} L(\theta) &= p(\vec{y} \mid X; \theta) \\ &= \prod_{i=1}^m p(y^{(i)} \mid x^{(i)}; \theta) \\ &= \prod_{i=1}^m (h_\theta(x^{(i)}))^{y^{(i)}} (1 - h_\theta(x^{(i)}))^{1-y^{(i)}} \end{aligned}$$

Logistic Regression

MLE

- It is easier to maximize the log likelihood: $p(y \mid x; \theta) = (h_\theta(x))^y (1 - h_\theta(x))^{1-y}$

$$\begin{aligned} L(\theta) &= p(\vec{y} \mid X; \theta) \\ &= \prod_{i=1}^m p(y^{(i)} \mid x^{(i)}; \theta) \\ &= \prod_{i=1}^m (h_\theta(x^{(i)}))^{y^{(i)}} (1 - h_\theta(x^{(i)}))^{1-y^{(i)}} \end{aligned}$$



$$\begin{aligned} \ell(\theta) &= \log L(\theta) \\ &= \sum_{i=1}^m y^{(i)} \log h(x^{(i)}) + (1 - y^{(i)}) \log(1 - h(x^{(i)})) \end{aligned}$$

Logistic Regression

$$p(y \mid x; \theta) = (h_{\theta}(x))^y (1 - h_{\theta}(x))^{1-y}$$

MLE Maximization

- How do we maximize the likelihood?

$$\begin{aligned}\ell(\theta) &= \log L(\theta) \\ &= \sum_{i=1}^m y^{(i)} \log h(x^{(i)}) + (1 - y^{(i)}) \log(1 - h(x^{(i)}))\end{aligned}$$

- Similar to our derivation in the case of linear regression, we can use gradient ascent.

Logistic Regression

MLE Recap

- In **logistic regression**, you predict the probability that a label y is 1 (versus 0) using:
$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$
- Then model the likelihood of each training example $(x^{(i)}, y^{(i)})$ as:
$$p(y^{(i)} | x^{(i)}; \theta) = (h_{\theta}(x^{(i)}))^{y^{(i)}} \cdot (1 - h_{\theta}(x^{(i)}))^{1-y^{(i)}}$$
- This tell us:
 - If $y^{(i)}=1$, the term becomes $h_{\theta}(x^{(i)})$.
 - If $y^{(i)}=0$, it becomes $1-h_{\theta}(x^{(i)})$.

2. Compute the likelihood over all examples

$$L(\theta) = \prod_{i=1}^m p(y^{(i)} | x^{(i)}; \theta) = \prod_{i=1}^m (h_{\theta}(x^{(i)}))^{y^{(i)}} \cdot (1 - h_{\theta}(x^{(i)}))^{1-y^{(i)}}$$

3. Use the Log-Likelihood

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^m \left[y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

4. Maximize via Gradient Ascent: to find the best θ , we maximize $\ell(\theta)$.

Logistic Regression

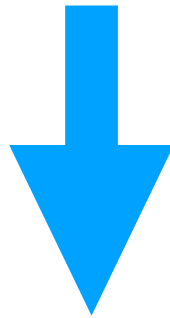
MLE Recap

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^m \left[y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

4. Maximize via Gradient Ascent: to find the best ϑ , we maximize $\ell(\vartheta)$.

a. Compute the gradient (derivative):

Especially focusing on one feature ϑ_j , the derivative with respect to ϑ_j is:


$$\frac{\partial \ell(\theta)}{\partial \theta_j} = (y - h_{\theta}(x)) \cdot x_j$$

b. Gradient Ascent Update Rule:

Since we want to maximize, we *add* a small step in the direction of the gradient:

where α is the learning rate.

$$\theta_j := \theta_j + \alpha \left(y^{(i)} - h_{\theta}(x^{(i)}) \right) x_j^{(i)}$$

Interestingly, this looks *just like* the update rule for least squares regression — the difference is in how $h_{\vartheta}(x)$ is defined (sigmoid vs. linear).

Logistic Regression

Compute the gradient (derivative)

We're trying to maximize this log-likelihood function:

$$\ell(\theta) = \sum_{i=1}^m \left[y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right]$$

$$\text{where } h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}.$$

We're Why Take a Derivative?

- The derivative tells us the slope: how much $\ell(\mathcal{J})$ changes if we change $\mathcal{J}$ (one parameter of $\mathcal{J}$).
- To find the maximum, we follow the slope (gradient ascent).

Logistic Regression

Compute the gradient (derivative)

Step 1: Focus on One Training Example Take just one example (x, y) . The log-likelihood contribution is:

$$\ell(\theta) = y \log(h_{\theta}(x)) + (1 - y) \log(1 - h_{\theta}(x)).$$

Step 2: Differentiate

We want $\frac{\partial \ell(\theta)}{\partial \theta_j}$

This is “how much does the log-likelihood change if I change θ_j ”?

Step 3: Use Chain Rule

Two facts help us:

1. Derivative of $\log(u)$ is $1/u$.

2. Derivative of the sigmoid is:

$$\frac{d}{dz} \sigma(z) = \sigma(z)(1 - \sigma(z)). \quad \text{Where } z = \theta^T x.$$

Logistic Regression

Compute the gradient (derivative)

Step 1: The log-likelihood contribution is: $\ell(\theta) = y \log(h_\theta(x)) + (1 - y) \log(1 - h_\theta(x))$.

Step 2: Differentiate $\frac{\partial \ell(\theta)}{\partial \theta_j}$

Step 3: Use Chain Rule $\frac{d}{dz} \sigma(z) = \sigma(z)(1 - \sigma(z))$.

Step 4: Plug the chain rule in log likelihood $\frac{\partial \ell(\theta)}{\partial \theta_j} = (y - h_\theta(x)) x_j$

Step 5: Intuition — If the true label y is **1** and the model predicted a small probability ($\hat{h}_\theta(x)=0.1$),

- Then $(y - \hat{h}_\theta(x)) = (1 - 0.1) = 0.9$.
- Gradient is positive $\rightarrow$ we *increase* θ_j to push probability higher.

If the true label y is **0** and the model predicted a high probability ($\hat{h}_\theta(x)=0.8$),

- Then $(y - \hat{h}_\theta(x)) = (0 - 0.8) = -0.8$.
- Gradient is negative $\rightarrow$ we *decrease* θ_j to push probability lower.

So the derivative naturally says:

“If the prediction is too low when $y=1$, push up. If too high when $y=0$, push down.”

Logistic Regression

Compute the gradient (derivative)

The gradient formula is so elegant:

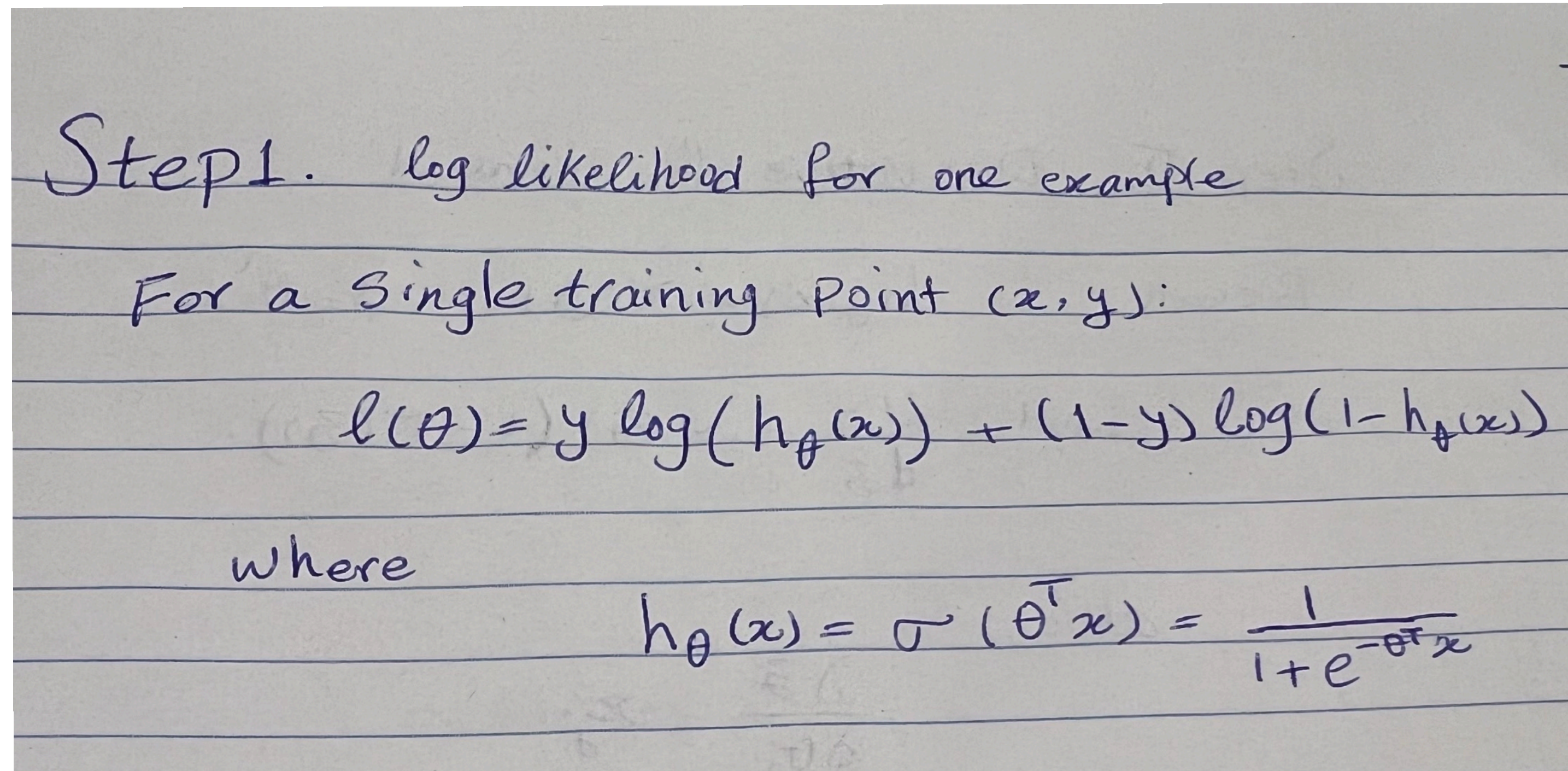
$$\frac{\partial \ell(\theta)}{\partial \theta_j} = (y - h_{\theta}(x)) x_j$$

It's just (error) $\times$ (input feature).

Logistic Regression — Self Study

Step by step Algebra

let's do the step-by-step algebra to show how we get



Step 1. log likelihood for one example

For a single training point (x, y) :

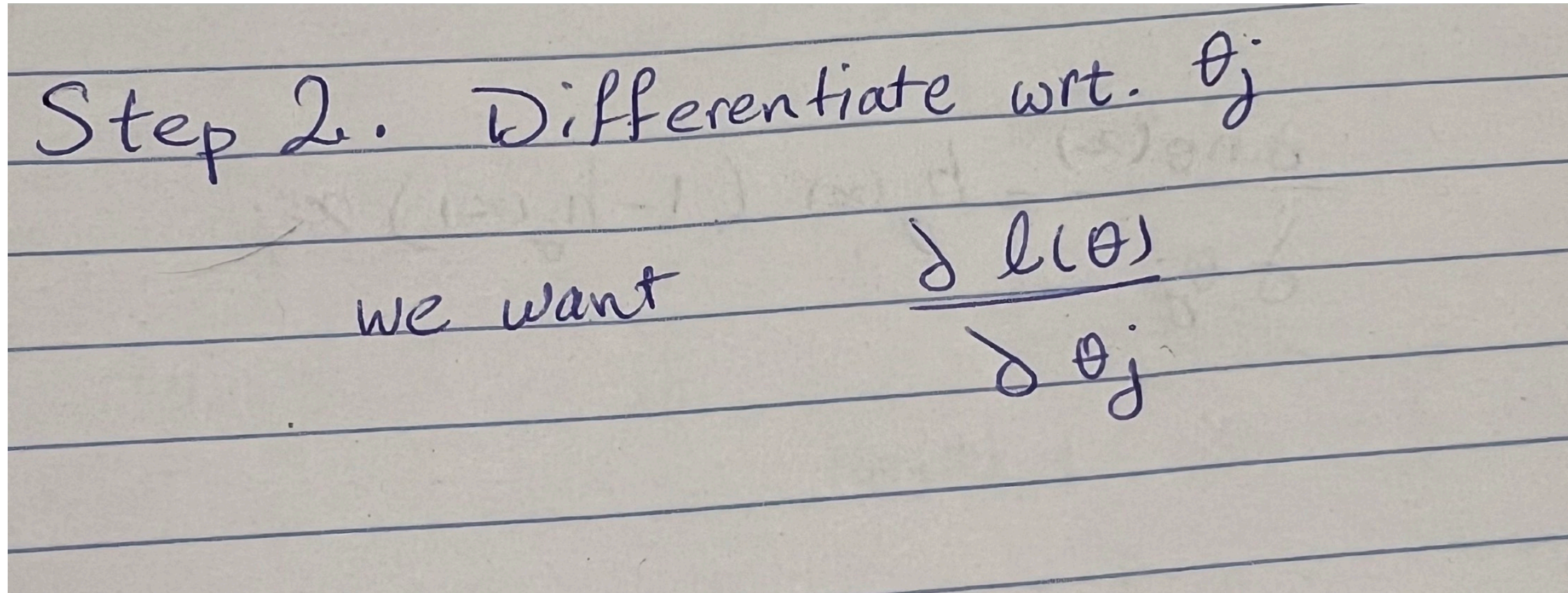
$$l(\theta) = y \log(h_{\theta}(x)) + (1-y) \log(1-h_{\theta}(x))$$

where

$$h_{\theta}(x) = \sigma(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

Logistic Regression — Self Study

Step by step Algebra



Logistic Regression – Self Study

Step by step Algebra

Step 3. Differentiate the two terms separately

$$\frac{\partial}{\partial \theta_j} [y \log(h_{\theta}(x))] = y \cdot \frac{1}{h_{\theta}(x)} \cdot \frac{\partial}{\partial \theta_j} h_{\theta}(x)$$
$$\frac{\partial}{\partial \theta_j} [(1-y) \log(1-h_{\theta}(x))] = (1-y) \cdot \frac{1}{1-h_{\theta}(x)} \cdot \left(-\frac{\partial h_{\theta}(x)}{\partial \theta_j}\right)$$

Logistic Regression – Self Study

Step by step Algebra

Step 4. Factor things, combine them:

$$\frac{\partial}{\partial \theta_j} l(\theta) = \left(\frac{y}{h_{\theta}(x)} - \frac{1-y}{1-h_{\theta}(x)} \right) \cdot \frac{\partial h_{\theta}(x)}{\partial \theta_j}$$

Logistic Regression — Self Study

Step by step

Step 5. Derivative of Sigmoid

Recall: $h_{\theta}(x) = \sigma(z)$, $z = \theta^T x$

and $\frac{d\sigma}{dz} = \sigma(z)(1 - \sigma(z))$.

also,

$$\frac{\partial z}{\partial \theta_j} = x_j$$

So,

$$\frac{\partial h_{\theta}(x)}{\partial \theta_j} = h_{\theta}(x)(1 - h_{\theta}(x))x_j$$

Logistic Regression – Self Study

Step by step Algebra

Step 6. Substitute back

$$\frac{\partial l(\theta)}{\partial \theta_j} = \left(\frac{y}{h_{\theta}(x)} - \frac{1-y}{1-h_{\theta}(x)} \right) \cdot [h_{\theta}(x)(1-h_{\theta}(x))x_j]$$

Logistic Regression – Self Study

Step by step Algebra

Step 7. Simplify the bracket

Compute the first part:

$$\frac{y}{h_{\theta}(x)} - \frac{1-y}{1-h_{\theta}(x)} = \frac{y(1-h_{\theta}(x)) - (1-y)h_{\theta}(x)}{h_{\theta}(x)(1-h_{\theta}(x))}$$

Simplify numerator:

$$= \frac{y - yh_{\theta}(x) - h_{\theta}(x) + yh_{\theta}(x)}{h_{\theta}(x)(1-h_{\theta}(x))}$$

$$= \frac{y - h_{\theta}(x)}{h_{\theta}(x)(1-h_{\theta}(x))}$$

Logistic Regression – Self Study

Step by step Algebra

Step 8. Cancel terms

Now plug back

$$\frac{\partial \ell(\theta)}{\partial \theta_j} = \frac{y - h_{\theta}(x)}{h_{\theta}(x)(1 - h_{\theta}(x))} \cdot [h_{\theta}(x)(1 - h_{\theta}(x))x_j]$$

the denominator and numerator cancel:

$$\frac{\partial \ell(\theta)}{\partial \theta_j} = (y - h_{\theta}(x))x_j$$

Logistic Regression — Self Study

Step by step Algebra

Final Result:

$$\frac{\partial l(\theta)}{\partial \theta_j} = (y - h_{\theta}(x)) x_j$$

Generalized Linear Model

GLM

Generalized Linear Models

Introduction

- So far, we've looked at two examples: one for regression and one for classification.
 - In regression, we assumed the output y (given x and θ) follows a **normal distribution**.
 - In classification, we assumed y follows a **Bernoulli distribution** (a coin flip).
- In both cases, the mean of that distribution (μ or ϕ) depends on x and the model's parameters θ .
- Now, we'll see that these are actually just two members of a bigger family of models called **Generalized Linear Models (GLMs)**.
- We'll also learn how other models in this family can be built and used for different regression and classification tasks.

Generalized Linear Models (GLM)

Introduction

- So far, we've looked at two examples: one for regression and one for classification.
 - In regression → predict continuous values
 - we assumed the output y (given x and β) follows a **normal distribution**.
 - In classification → predict 0/1 categories
 - we assumed y follows a **Bernoulli distribution** (a coin flip).
- In both cases, the mean of that distribution (μ or ϕ) depends on x and the model's parameters β .
- **Generalized Linear Models (GLMs)**
 - Choose a **probability distribution** for the output y (Normal for regression, Bernoulli for classification, Poisson for counts, etc.).
 - **Link function** to allow the use of a linear combination of predictors (e.g., $\beta_0 + \beta_1 X$) to model the expected outcome.

Generalized Linear Models (GLM)

Why?

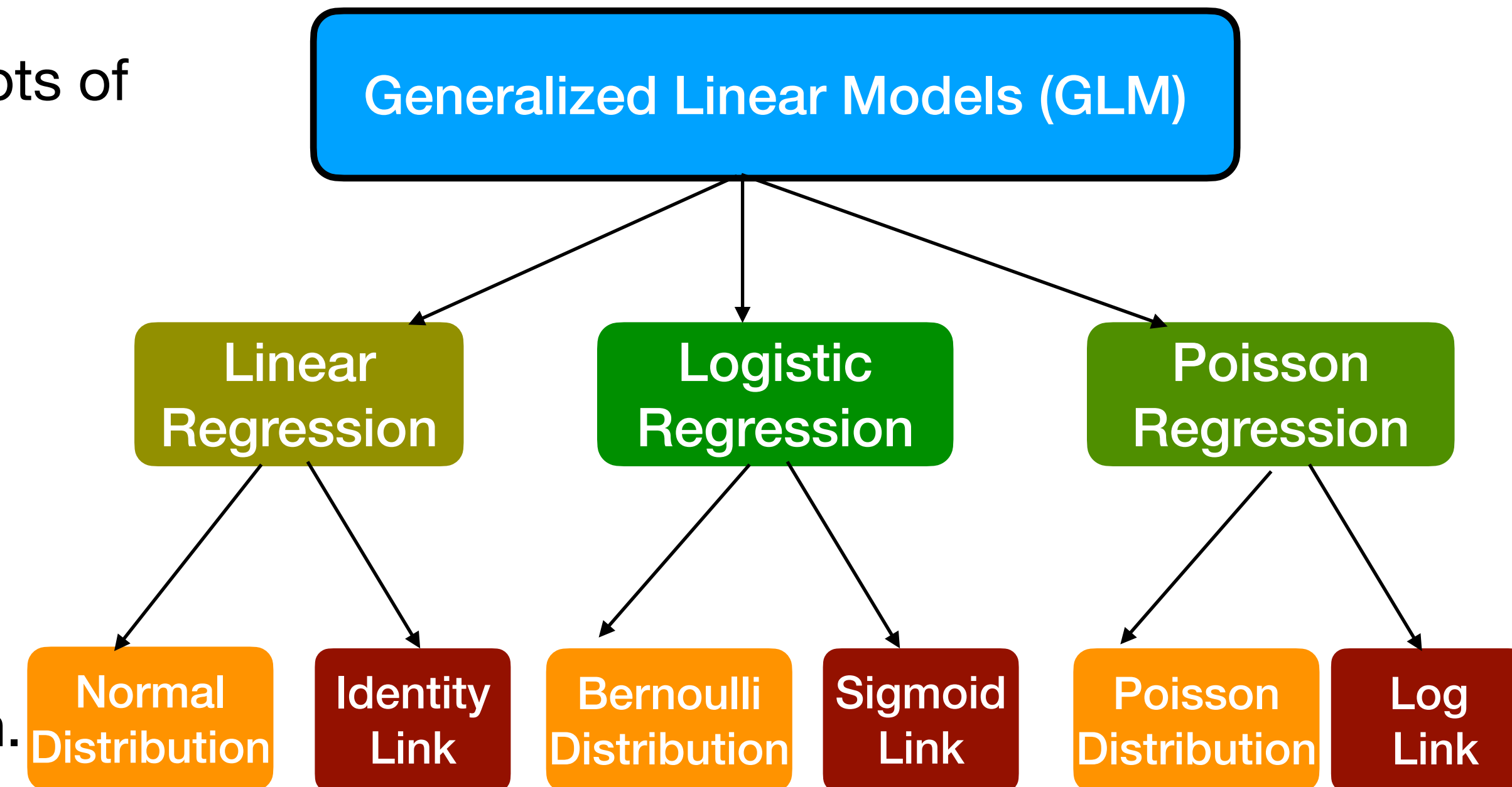
- GLMs justify why we:
 - Use least squares in linear regression (comes from Normal likelihood). $y \mid x; \theta \sim \mathcal{N}(\mu, \sigma^2)$
 - Use the sigmoid/log-likelihood in logistic regression (comes from Bernoulli likelihood).
 $y \mid x; \theta \sim \text{Bernoulli}(\phi),$
- Without GLMs, these formulas might feel arbitrary.
- With GLMs, you see they all come from the same statistical principle: **maximum likelihood estimation under a chosen distribution.**

Generalized Linear Models (GLM)

many other supervised models

- Once you see the GLM recipe, you realize you can create lots of models:

- **Poisson regression** for count data (e.g., number of hospital visits).
- **Exponential regression** for survival times.
- **Multinomial regression** for multi-class classification.



- So instead of memorizing separate models, you learn one “template” (GLM) and just plug in the distribution + link function you need.

GLM: The Exponential Family

Class of distributions

- **Definition of Exponential family distributions:** We say that a class of distributions is in the exponential family if it can be written in the form:

$$p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$$

GLM: The Exponential Family

Class of distributions

- **Definition of Exponential family distributions:** We say that a class of distributions is in the exponential family if it can be written in the form:

$$p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$$

The diagram illustrates the components of the exponential family distribution formula. Red circles highlight the natural parameter η , the base measure $b(y)$, and the log partition function $a(\eta)$. Arrows point from these circles to blue boxes below. An arrow also points from the exponent term $\eta^T T(y)$ to a box.

Component	Label	Equation
η	natural parameter canonical parameter	
$b(y)$	Base measure	
$\eta^T T(y)$	sufficient statistic	$T(y) = y$
$a(\eta)$	log partition function	$T(y) = y$

GLM: The Exponential Family

Class of distributions

- **Definition of Exponential family distributions:** We say that a class of distributions is in the exponential family if it can be written in the form:

$$p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$$

- A fixed choice of $\mathcal{T}$, a and b defines a family (or set) of distributions that is parameterized by η ;
- as we vary η , we then get different distributions within this family.

.

GLM: The Exponential Family

Class of distributions

- **Definition of Exponential family distributions:** We say that a class of distributions is in the exponential family if it can be written in the form:

$$p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$$

- We now show that the **Bernoulli** and the **Gaussian** distributions are examples of exponential family distributions.

GLM: The Exponential Family

Gaussian Distribution

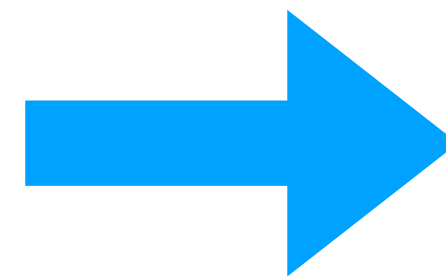
Gaussian (Normal) with mean μ , variance σ^2 :

$$p(y; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(y - \mu)^2\right)$$

Expand the quadratic in the exponent:

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2\right) \cdot \exp\left(\mu y - \frac{1}{2}\mu^2\right)$$

$$p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$$



$$\begin{aligned}\eta &= \mu \\ T(y) &= y \\ a(\eta) &= \mu^2/2 \\ &= \eta^2/2 \\ b(y) &= (1/\sqrt{2\pi}) \exp(-y^2/2)\end{aligned}$$

GLM: The Exponential Family

Bernoulli Distribution

The Bernoulli distribution with mean ϕ , specifies a distribution over $y \in \{0, 1\}$

$$p(y = 1; \phi) = \phi; \quad p(y = 0; \phi) = 1 - \phi.$$

As we vary ϕ , we obtain Bernoulli distributions with different means. We now show that this class of Bernoulli distributions, ones obtained by varying ϕ , is in the exponential family.

$$\begin{aligned} p(y; \phi) &= \phi^y (1 - \phi)^{1-y} \\ &= \exp(y \log \phi + (1 - y) \log(1 - \phi)) \\ &= \exp \left(\left(\log \left(\frac{\phi}{1 - \phi} \right) \right) y + \log(1 - \phi) \right) \end{aligned}$$

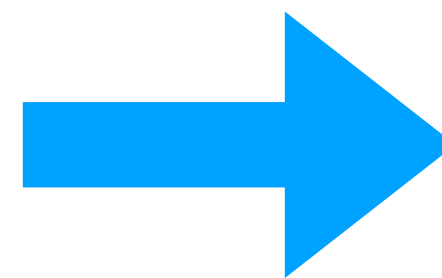
GLM: The Exponential Family

Bernoulli Distribution

$$\begin{aligned}p(y; \phi) &= \phi^y (1 - \phi)^{1-y} \\&= \exp(y \log \phi + (1 - y) \log(1 - \phi)) \\&= \exp \left(\left(\log \left(\frac{\phi}{1 - \phi} \right) \right) y + \log(1 - \phi) \right)\end{aligned}$$

the natural parameter is given by: $\eta = \log(\phi / (1 - \phi))$

$$p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$$



$$\begin{aligned}T(y) &= y \\a(\eta) &= -\log(1 - \phi) \\&= \log(1 + e^\eta) \\b(y) &= 1\end{aligned}$$

Constructing GLMs

Constructing GLMs $p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$

Example

Modeling shop/store visitor:

y (**target**): the number of customers arriving in your store

x (**features**): store promotions, recent advertising, weather, day-of-week, etc.

Distribution: Poisson distribution usually gives a good model for numbers of visitors.

how can we come up with a model for our problem?

Constructing GLMs $p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$

GLM: Store Visitors

how can we come up with a model for our problem?

Modeling shop/store visitor:

y (**target**): the number of customers arriving in your store

x (**features**): store promotions, recent advertising, weather, day-of-week, etc.

Distribution: Poisson distribution

To derive a GLM for this problem, we will make the following three assumptions about the conditional distribution of y given x and about our model.

1. $y | x, \theta \sim \text{ExponentialFamily}(\eta)$.

Assumption 1: The outputs y come from an exponential family distribution.

Assumption 2: The mean of y depends on x through a linear function of the inputs.

Assumption 3: There exists a link function that connects the mean of y to that linear predictor.

Constructing GLMs $p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$

GLM: Store Visitors

Modeling shop/store visitor:

y (**target**): the number of customers arriving in your store

x (**features**): store promotions, recent advertising, weather, day-of-week, etc.

Distribution: Poisson distribution

Assumption 1: The outputs y come from an exponential family distribution.

- We assume that, given inputs x , the target y follows some distribution from the **exponential family**.
- Examples:
 - Gaussian (for continuous values)
 - Bernoulli (for binary values)
 - Poisson (for counts)
- This just means we restrict ourselves to a “nice family” of distributions that are mathematically convenient and cover most practical cases.

$$1. y \mid x; \theta \sim \text{ExponentialFamily}(\eta).$$

In other words: *“Pick a probability distribution for your data that belongs to the exponential family.”*

Constructing GLMs

Family of Exponential Distributions

Continuous Distributions

- Normal (Gaussian) distribution
- Exponential distribution
- Gamma distribution
- Chi-squared distribution
- Beta distribution
- Inverse Gaussian distribution
- Log-normal distribution

Discrete Distributions

- Bernoulli distribution
- Binomial distribution
- Categorical distribution
- Poisson distribution

Multivariate Distributions

- Dirichlet distribution
- Wishart distribution
- Multivariate Normal distribution

Constructing GLMs $p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$

GLM: Store Visitors

Modeling shop/store visitor:

y (**target**): the number of customers arriving in your store

x (**features**): store promotions, recent advertising, weather, day-of-week, etc.

Distribution: Poisson distribution

Assumption 2: The mean of y depends on x through a linear function of the inputs.

- We introduce a **linear predictor**: $\eta = \mathcal{W}^T x$
- This doesn't mean y itself is linear in x . It means some function of its expected value will be linear in x .

In other words: “*The model combines the input features linearly (like in regression).*”

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Assumption 3: There exists a link function that connects the mean of y to that linear predictor.

- The mean of y (expected value) is connected to the linear predictor via a **link function** g :

$$g(\mathbb{E}[y \mid x]) = \theta^T x$$

- Different choices of link function $\rightarrow$ different models:
 - Identity link $g(\mu)=\mu \rightarrow$ Linear Regression.
 - Logit link $g(\mu)=\log(\mu/(1-\mu)) \rightarrow$ Logistic Regression.
 - Logit link $g(\mu)=\log(\mu) \rightarrow$ Poisson Regression.

In other words: “*We don’t force the mean of y itself to be linear in x . Instead, we let a function of the mean be linear in x .*”

Constructing GLMs $p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$

Putting it all together

The three assumptions say:

1. Pick a distribution for y from the exponential family.
2. Combine inputs x linearly with parameters η .
3. Use a link function to connect that linear combination to the mean of y .

And that's the recipe for a **Generalized Linear Model (GLM)**.



Let's take 15 minutes break!

Naïve Bayes

Naive Bayes Classifier

Naive Bayes is a probabilistic machine-learning classifier that assigns a class to a data point based on **likelihoods**.

It makes the simplifying assumption that all features contribute **independently** to the prediction.

The main idea behind the Naive Bayes classifier is to use **Bayes' Theorem** to classify data based on the probabilities of different classes given the features of the data.

Bayes' Theorem

Bayes' Theorem is a mathematical formula used to determine the conditional probability of an event based on prior knowledge and new evidence.

It tells you how to revise the chance of something being true when you receive new evidence. It combines two key ideas:

- **Prior belief (prior probability):** what you thought the probability was *before* seeing new information.
- **Evidence (new data):** the observation you just made that should influence your belief.

Bayes' Theorem

Bayes' Theorem is a mathematical formula used to determine the conditional probability of an event based on prior knowledge and new evidence.

Bayes Theorem formula: For any two events A and B, Bayes's formula for the Bayes theorem is given by:

The diagram shows the Bayes' Theorem formula:
$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$
 The formula is presented on a light gray rectangular background. Red arrows point from the text labels to specific parts of the formula: one arrow points from 'Posterior Probability' to $P(A|B)$, another from the same label to the fraction bar, and a third from 'Prior Probability' to $P(A)$.

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Posterior Probability

Prior Probability

Assumptions of Naive Bayes

The fundamental Naive Bayes assumption is that each feature makes an:

- **Feature independence:** This means that when we are trying to classify something, we assume that each feature (or piece of information) in the data does not affect any other feature.
- **Continuous features are normally distributed:** If a feature is continuous, then it is assumed to be normally distributed within each class.
- **Discrete features have multinomial distributions:** If a feature is discrete, then it is assumed to have a multinomial distribution within each class.
- **Features are equally important:** All features are assumed to contribute equally to the prediction of the class label.
- **No missing data:** The data should not contain any missing values.

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$$P(x_1, x_2, \dots, x_n | y) = P(x_1 | y) \cdot P(x_2 | y) \cdots P(x_n | y)$$

Classification Task

- Given a data set of samples represented by 3 features $\{x_1, x_2, x_3\}$ we want to predict the class label $y = P(x_1, x_2, x_3 | y)$

Classification Task

- Given a data set of samples represented by 3 features $\{x_1, x_2, x_3\}$ we want to predict the class label $y = P(x_1, x_2, x_3 | y)$?
- We must use the **Chain Rule of Probability**
- To calculate $P(x_1, x_2, x_3 | y)$, you must account for the dependencies between features by conditioning each subsequent feature on all those that preceded it.

The Chain Rule Formula

$$P(x_1, x_2, x_3 | y) = P(x_1 | y) \cdot P(x_2 | x_1, y) \cdot P(x_3 | x_1, x_2, y)$$

The problem of Chain Rule in ML

- Why is this rarely used in ML?

$$P(x_1, x_2, x_3|y) = P(x_1|y) \cdot P(x_2|x_1, y) \cdot P(x_3|x_1, x_2, y)$$

The problem of Chain Rule in ML

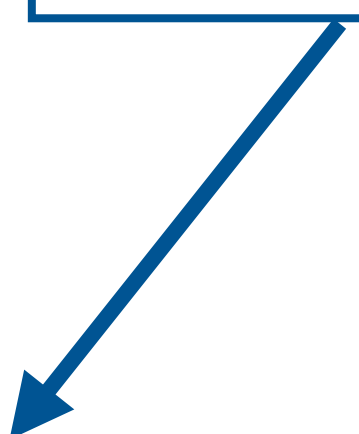
- Why is this rarely used in ML?
 - **Curse of Dimensionality:** As you add more features, the number of combinations grows exponentially.
 - If even one combination is missing from your data, the probability defaults to zero.

$$P(x_1, x_2, x_3|y) = P(x_1|y) \cdot P(x_2|x_1, y) \cdot P(x_3|x_1, x_2, y)$$

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$$P(x_1, x_2, \dots, x_n|y) = P(x_1|y) \cdot P(x_2|y) \cdots P(x_n|y)$$

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$$P(x_1, x_2, \dots, x_n | y) = P(x_1 | y) \cdot P(x_2 | y) \cdots P(x_n | y)$$

Therefore, the Bayes' theorem becomes:

$$P(y | x_1, \dots, x_n) = \frac{P(y) \cdot \prod_{i=1}^n P(x_i | y)}{P(X)}$$

Assumptions of Naive Bayes

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$$P(x_1, x_2, \dots, x_n | y) = P(x_1 | y) \cdot P(x_2 | y) \cdots P(x_n | y)$$

$$P(y | x_1, \dots, x_n) = \frac{P(y) \cdot \prod_{i=1}^n P(x_i | y)}{P(X)}$$

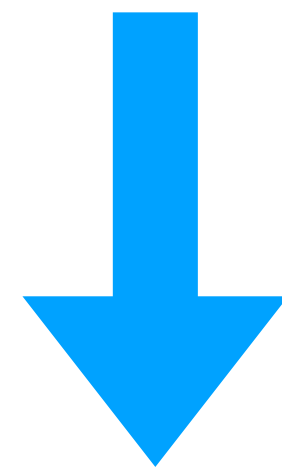
Because the denominator is constant for a given input, we can write:

$$P(y | x_1, \dots, x_n) \propto P(y) \cdot \prod_{i=1}^n P(x_i | y)$$

Assumptions of Naive Bayes

To build a Naive Bayes Classifier we need to compute the posterior for each class y and choose the class with the highest probability:

$$P(y|x_1, \dots, x_n) \propto P(y) \cdot \prod_{i=1}^n P(x_i|y)$$



Number of training samples

$$\hat{y} = \arg \max_y P(y) \cdot \prod_{i=1}^n P(x_i|y)$$

Predicted class

Class label in training sample

Example of Naive Bayes

Predicting if an accident can happen?

Feature	Class/labels
Weather condition	rain,snow,clear
Road condition	good, normal, bad
Traffic condition	normal, high, light
Engine problem	yes,no

SNo.	Weather condition	Road condition	Traffic condition	Engine problem	Accident
1	Rain	bad	high	no	yes
2	snow	average	normal	yes	yes
3	clear	bad	light	no	no
4	clear	good	light	yes	yes
5	snow	good	normal	no	no
6	rain	average	light	no	no
7	rain	good	normal	no	no
8	snow	bad	high	no	yes
9	clear	good	high	yes	no
10	clear	bad	high	yes	yes

[Example resource](#)

Example of Naive Bayes

Suppose we are now given a new feature vector:

Weather condition: rain

Road condition: good

Traffic condition: normal

Engine problem: no

The task is to predict if an accident will happen?

$$\hat{y} = \arg \max_y P(y) \cdot \prod_{i=1}^n P(x_i|y)$$

SNo.	Weather condition	Road condition	Traffic condition	Engine problem	Accident
1	Rain	bad	high	no	yes
2	snow	average	normal	yes	yes
3	clear	bad	light	no	no
4	clear	good	light	yes	yes
5	snow	good	normal	no	no
6	rain	average	light	no	no
7	rain	good	normal	no	no
8	snow	bad	high	no	yes
9	clear	good	high	yes	no
10	clear	bad	high	yes	yes

Feature	Class/labels
Weather condition	rain,snow,clear
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[Example resource](#)

Types of Naive Bayes

$$\hat{y} = \arg \max_y P(y) \cdot \prod_{i=1}^n P(x_i|y)$$

1. **Gaussian Naive Bayes:** feature are assumed to be distributed according to a Gaussian distribution
2. **Multinomial Naive Bayes:** features represent the frequency of terms (such as word counts) in a document.
3. **Bernoulli Naive Bayes:** deals with binary features, where each feature indicates whether a word appears or not in a document.

MLE vs MAP

Maximum Likelihood Estimation (MLE)

Maximum A Posteriori (MAP)

MLE vs MAP

- To generalize the Bayes approach, we can imagine each dataset can be represented with a probability distribution.
- Each probability distribution is a model with its parameter. Our job is to find those parameter.
- Suppose we are given a dataset $D = \{x_1, x_2, \dots, x_n\}$ which we assume is generated from a probabilistic model parameterized by θ .
- Our goal is to estimate the unknown parameter θ using the observed data.
- Let $(x | \theta)$ denote the likelihood function, i.e., the probability of observing data x given parameter θ .

MLE vs MAP

Maximum Likelihood Estimation (MLE)

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(D \mid \theta)$$

Maximum A Posteriori (MAP)

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid D)$$

MLE vs MAP

Maximum Likelihood Estimation (MLE)

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(D \mid \theta)$$

- Pick the parameter θ that makes the observed data D most likely.
- It uses only the data.
- No “prior belief” about θ .

Maximum A Posteriori (MAP)

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid D)$$

MLE vs MAP

Maximum Likelihood Estimation (MLE)

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(D \mid \theta)$$

Maximum A Posteriori (MAP)

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid D)$$

- “Pick the parameter θ that is most probable *after seeing the data*.”
- This is **Bayesian**: it combines data +
- And it also adds prior belief

MLE vs MAP

Maximum Likelihood Estimation (MLE)

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(D \mid \theta)$$

Assuming that the data are independent and identically distributed (i.i.d.)

$$p(D \mid \theta) = \prod_{i=1}^n p(x_i \mid \theta)$$

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \log p(D \mid \theta) = \arg \max_{\theta} \sum_{i=1}^n \log p(x_i \mid \theta)$$

MLE vs MAP

Maximum Likelihood Estimation (MLE)

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \log p(D \mid \theta) = \arg \max_{\theta} \sum_{i=1}^n \log p(x_i \mid \theta)$$

Maximum A Posteriori (MAP)

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid D)$$

Using Bayes Theorem

$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{p(D)}$$

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(D \mid \theta)p(\theta)$$

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} [\log p(D \mid \theta) + \log p(\theta)]$$

MLE vs MAP

Maximum Likelihood Estimation (MLE)

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \log p(D \mid \theta) = \arg \max_{\theta} \sum_{i=1}^n \log p(x_i \mid \theta)$$

Maximum A Posteriori (MAP)

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} [\log p(D \mid \theta) + \log p(\theta)]$$

When to Use MLE vs MAP

Maximum Likelihood Estimation (MLE)

- You have a large dataset, and the effect of prior beliefs is negligible.
- You lack reliable prior information.
- You aim for models grounded in frequentist statistics.

Maximum A Posteriori (MAP)

- You have small or noisy datasets.
- You possess strong prior knowledge or domain expertise.
- You want to regularize your estimate to avoid overfitting.

Assignment

1. Form Your Group

Send me your group by Thursday September 4th at 5:00 pm PST.

2. Work on your Proposal

Submit your proposal by Saturday September 14th at 6:00 pm PST.

Only one submission per group.



Questions?

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